# Learning Curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Learning Curves before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Overfitting

Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

Complete the supported Learning Curves trace and record the first step you can explain without help.

Use W2: Read both curves for **Learning curve**s and **variability**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Learning Curves explanation against the supplied facts before discussing it.

Use the worked case for **Learning curve**s and **variability**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

**Independent case: Apply independently**

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Learning Curves target to a trace and a justified conclusion.

**Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Learning Curves case and show the deciding evidence.

With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

Interpret the curve and uncertainty.

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

## Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

As **training** examples grow from 50 to 100 to 200, validation accuracy is **60%**, 72%, 78%. What does the curve suggest, and what does it not guarantee?

My independent answer and reason

**Write or draw your reasoning and evidence.**

**Week 28 Test and reflect**

WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

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## Project checkpoint

> Justify what Learning Curves adds to Responsible Plant Classifier and keep the evidence.

Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

**Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

## W1 Summarize repeats

> Summarize repeats. Apply the supplied Learning Curves facts and keep your reasoning.

Calculate the mean and **range** for the 20-row condition.

**My reasoning and evidence**

WORKBOOK · 13 OF 21

## W2 Read both curves

> Read both curves. Apply the supplied Learning Curves facts and keep your reasoning.

Why can **training** error rise while validation error falls?

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Avoid extrapolation

Avoid extrapolation. Apply the supplied Learning Curves facts and keep your reasoning.

Can the 40-row result guarantee a mean MAE of 2.5 at 80 rows?

**My reasoning and evidence**

WORKBOOK · 15 OF 21

## W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Learning Curves facts and keep your reasoning.

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

**My reasoning and evidence**

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WORKBOOK · 16 OF 21

## W5 Explain the spread

Explain the spread. Apply the supplied Learning Curves facts and keep your reasoning.

Why is an observed **range** not a confidence interval?

**My reasoning and evidence**

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WORKBOOK · 17 OF 21

## Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 18 OF 21**

## Pictorial case practice: Learning Curves

> Read a learning curve with repeated-run spread rather than drawing a certainty arrow. Explain the labelled step before checking the changed condition.

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/28

# Learning Curves

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



**1. Summarise first size**

Mean6 errors; range5-7, corresponding to mean 40% accuracy.





**2. Summarise larger size**

Mean4 errors; range3-5, corresponding to mean 60% accuracy.





**3. State bounded pattern**

More training plants improved the average in these supplied runs; variability remains.

**CHANGE ONE THING**

A30-plant experiment gives errors [4, 4, 4].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? More **training** plants improved the average in these supplied runs; **variability** remains.

1. State bounded pattern

2. Summarise larger size

3. Summarise first size

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: State bounded pattern More **training** plants improved the average in these supplied runs; **variability** remains. Read the picture in order.

More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: A30-plant experiment gives errors [4, 4, 4]. Which conclusion is supported by the supplied facts?

1. Mean remains4; this additional size does not improve the observed mean over20 plants.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Mean remains4; this additional size does not improve the observed mean over20 plants.

**Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

### Repair a missing building block

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

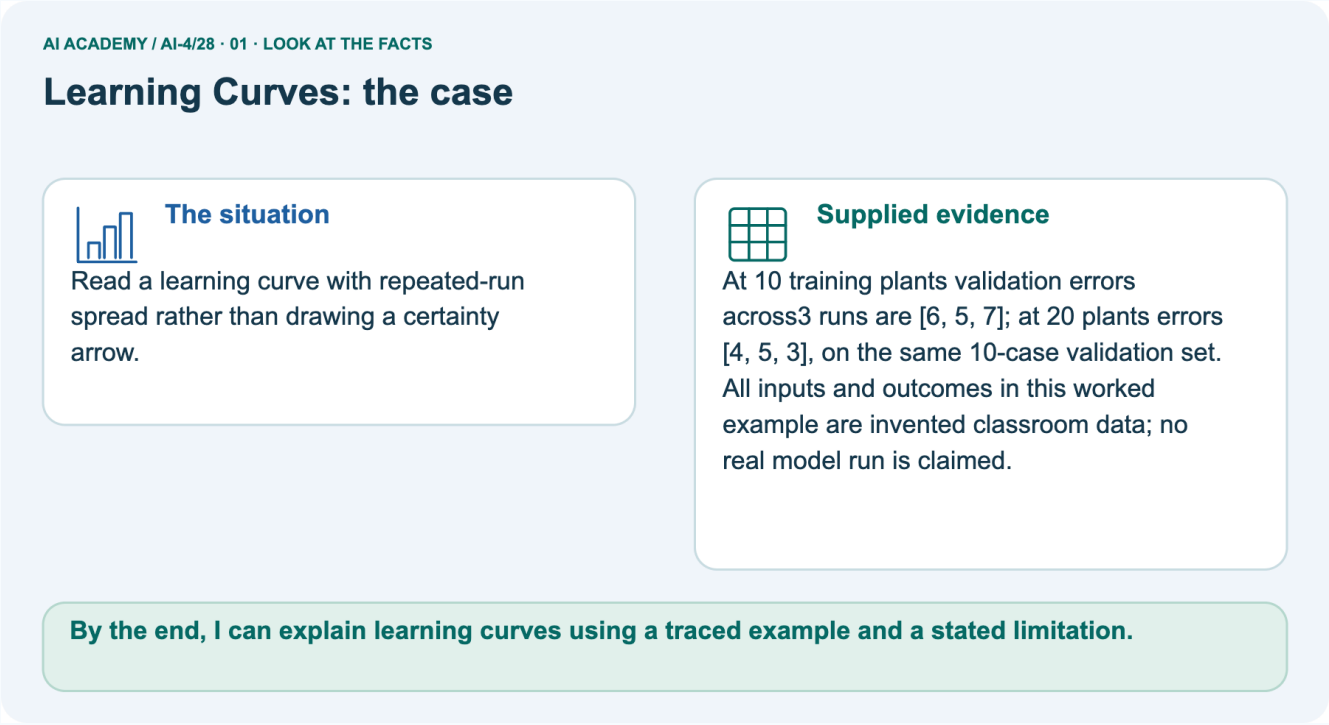
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each Learning Curves picture from its labels, then record a relevant limit.

### 01. Learning Curves: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

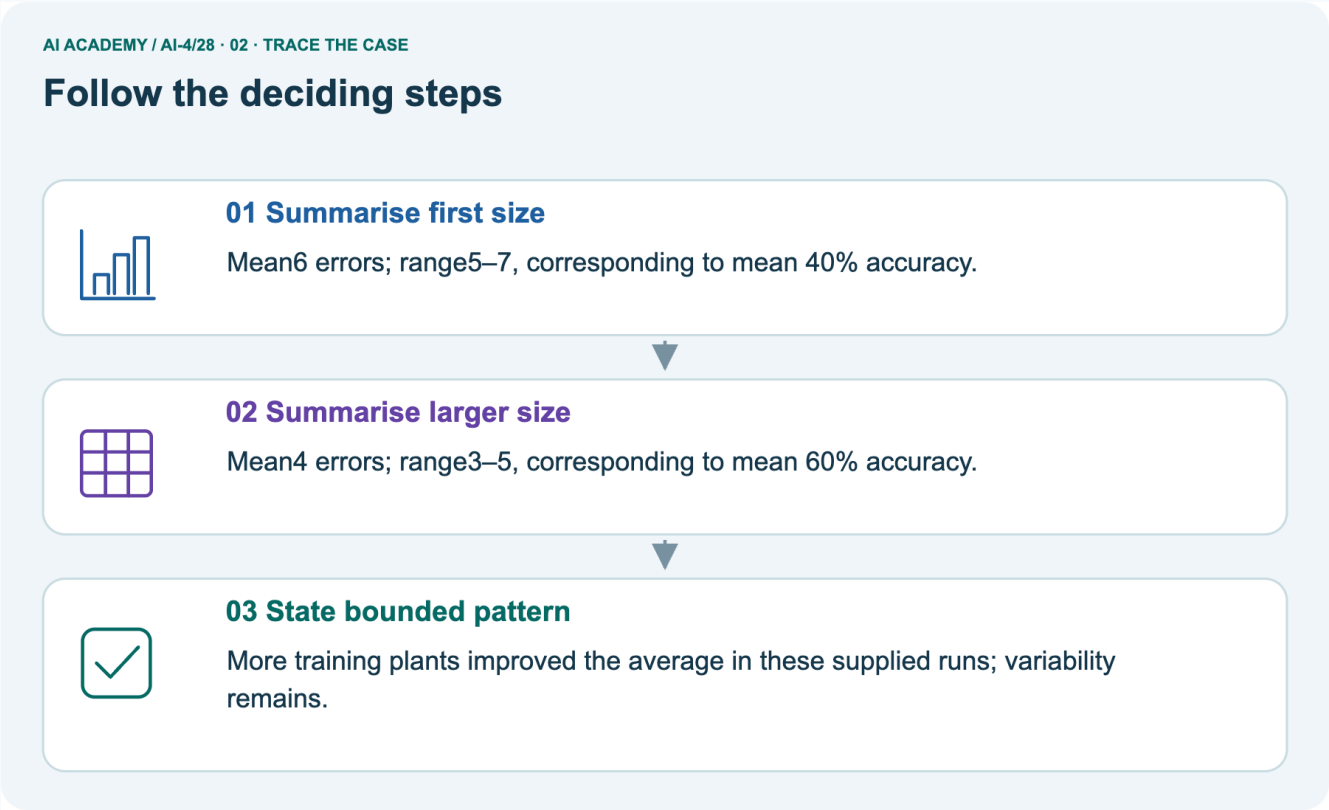
Supplied facts: At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



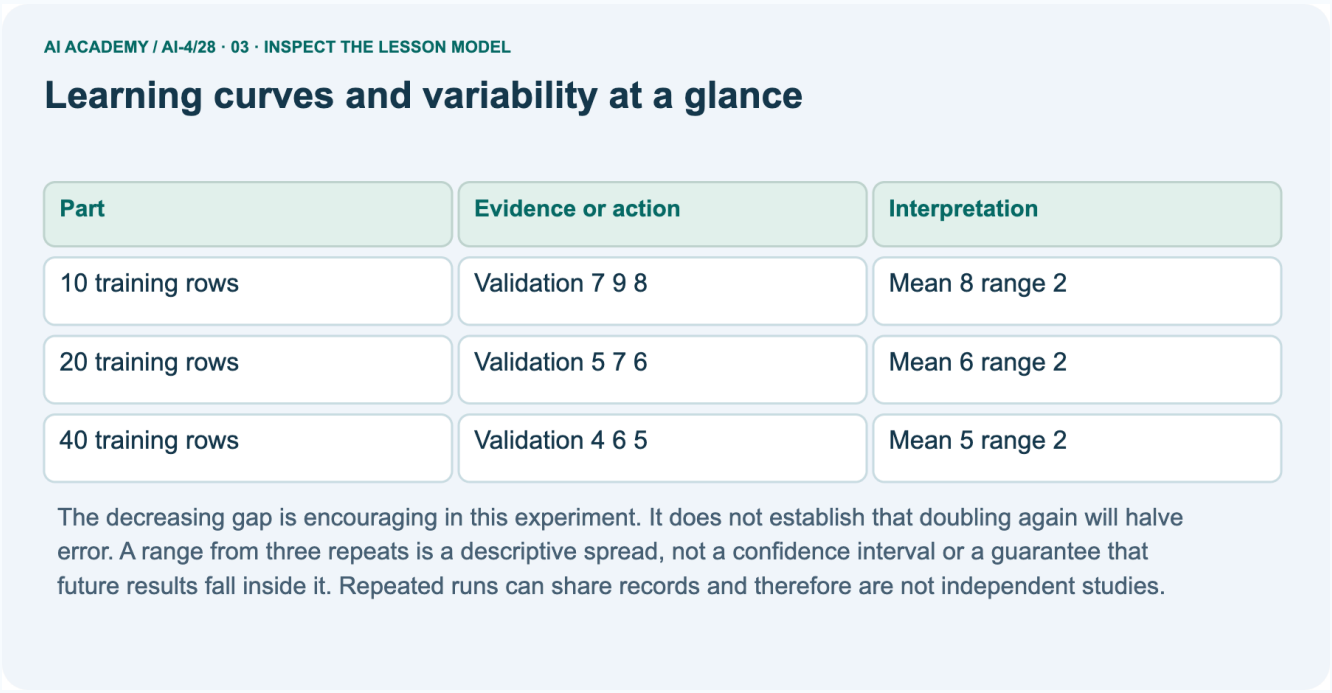

Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise first size: Mean6 errors; **range**5–7, corresponding to mean 40% accuracy. → Summarise larger size: Mean4 errors; **range**3–5, corresponding to mean **60%** accuracy. → State bounded pattern: More **training** plants improved the average in these supplied runs; **variability** remains.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

## 03. Learning curves and variability at a glance



AI ACADEMY / AI-4/28 · 03 · INSPECT THE LESSON MODEL

### Learning curves and variability at a glance

| Part | Evidence or action | Interpretation |
|---|---|---|
| 10 training rows | Validation 7 9 8 | Mean 8 range 2 |
| 20 training rows | Validation 5 7 6 | Mean 6 range 2 |
| 40 training rows | Validation 4 6 5 | Mean 5 range 2 |

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

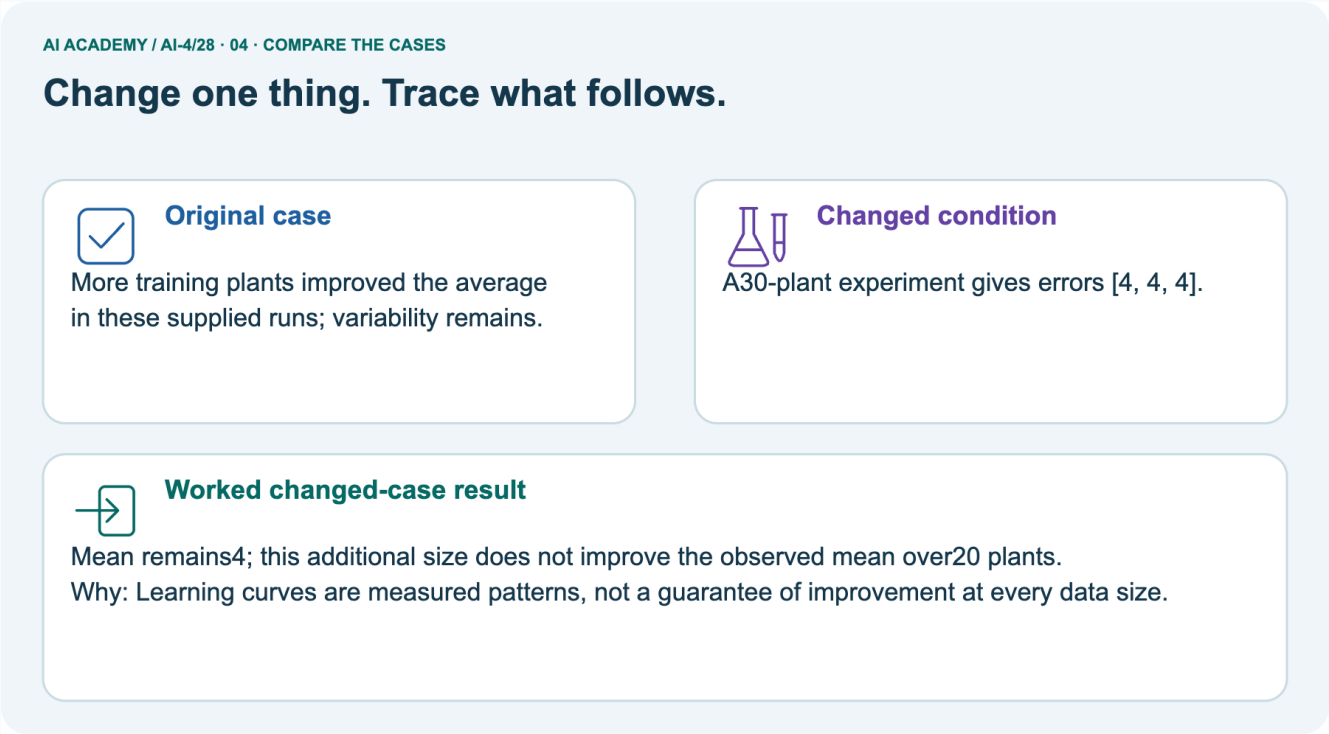
Follow the labelled picture with the matching explanation. Open the vector diagram

**Learning curve**s and **variability** at a glance. 10 **training** rows | Validation 7 9 8 | Mean 8 **range** 2. 20 **training** rows | Validation 5 7 6 | Mean 6 **range** 2. 40 **training** rows | Validation 4 6 5 | Mean 5 **range** 2

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

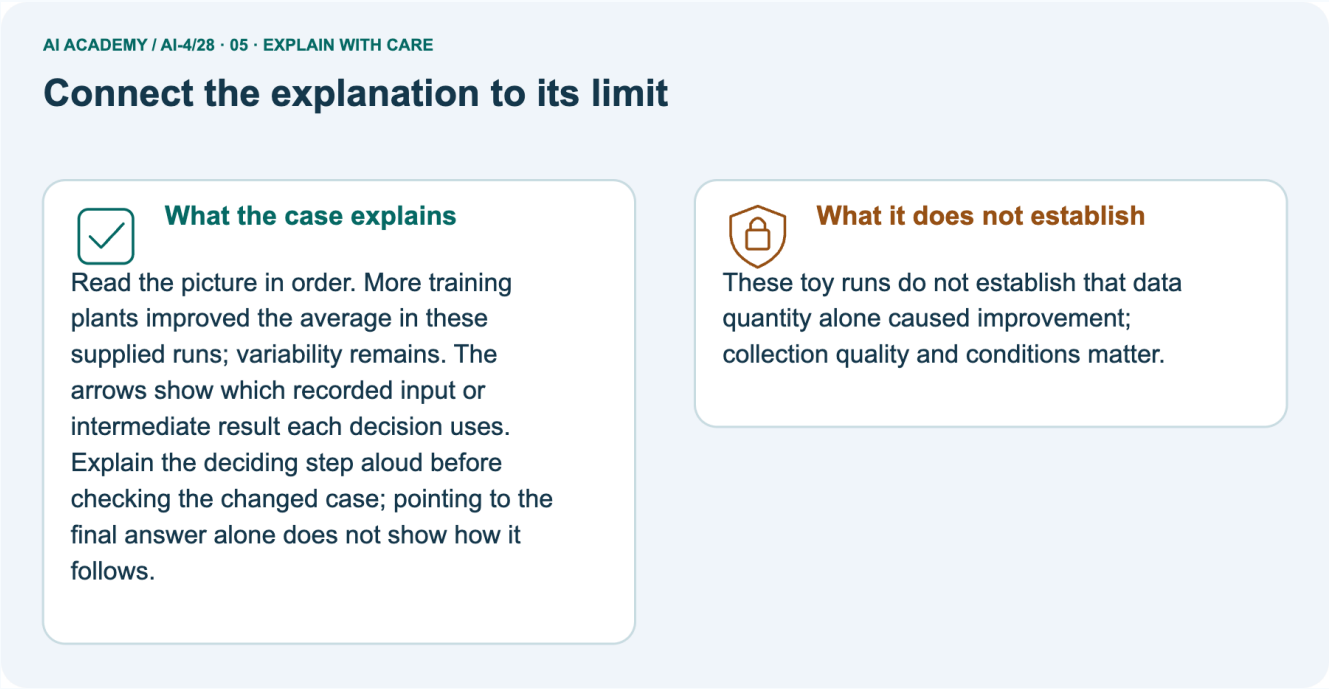
Original case: More **training** plants improved the average in these supplied runs; **variability** remains. Changed condition: A30-plant experiment gives errors [4, 4, 4].

Expected result: Mean remains4; this additional size does not improve the observed mean over20 plants. Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

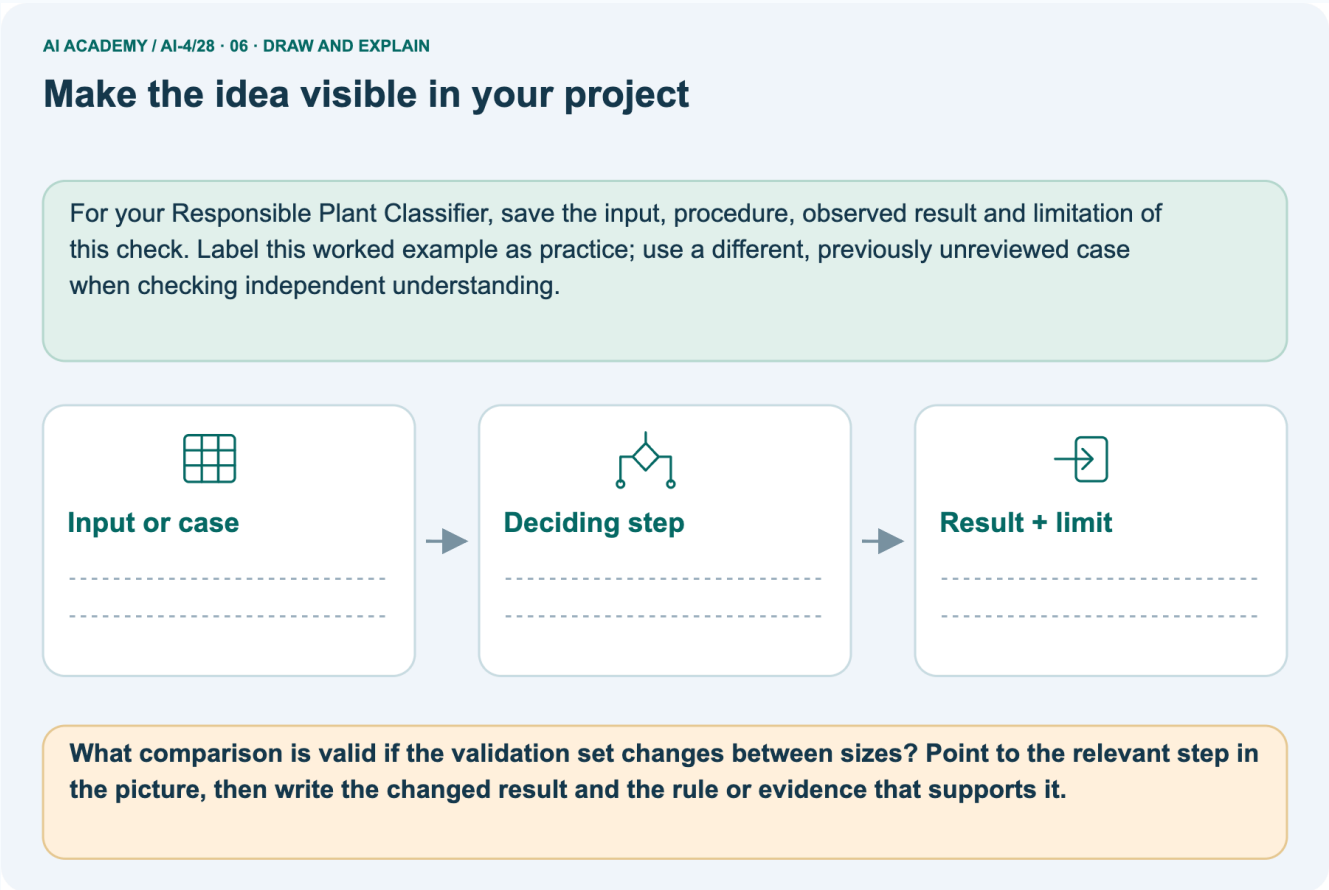
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.